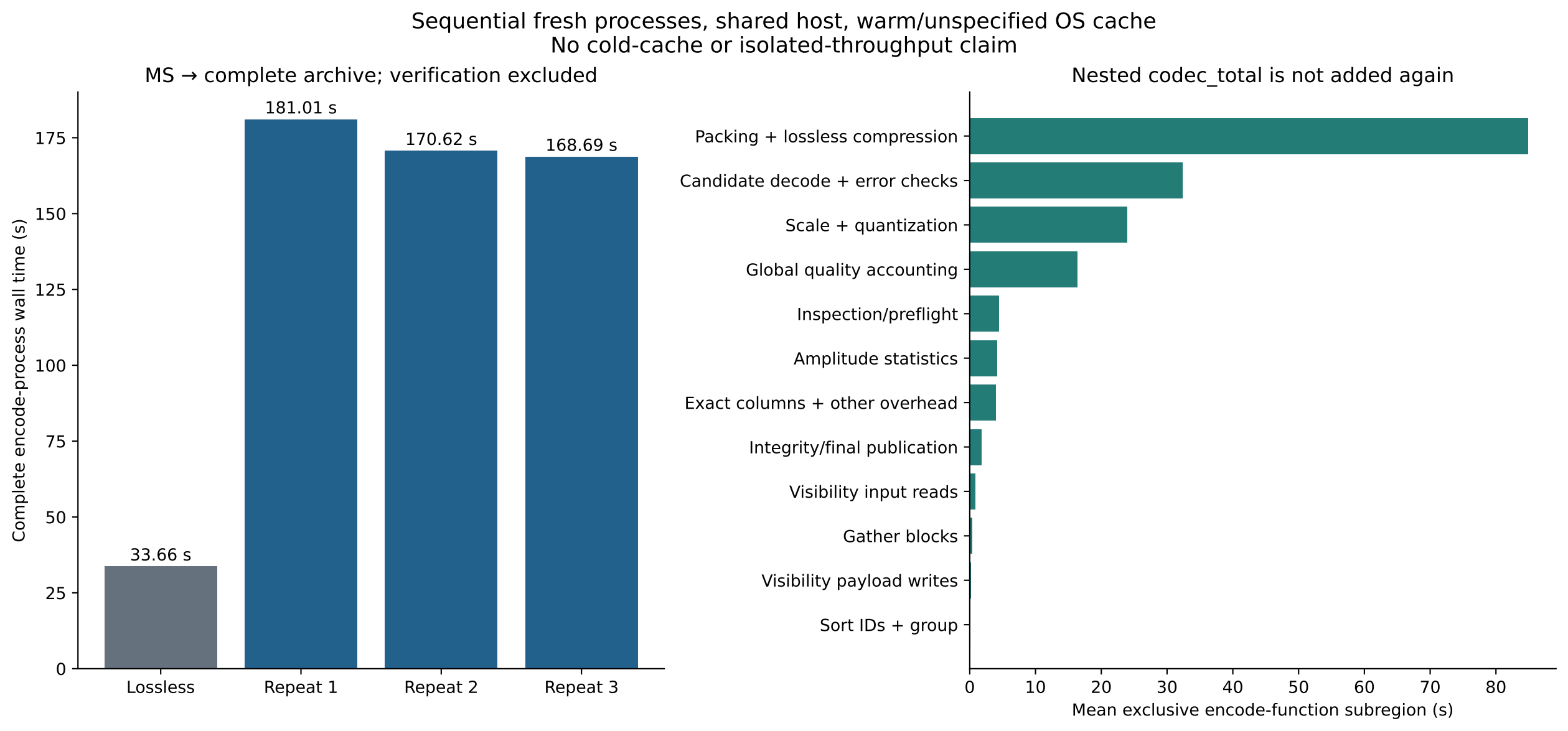
Sequential fresh processes, shared host, warm/unspecified OS cache
No cold-cache or isolated-throughput claim
MS → complete archive; verification excluded
Nested codec_total is not added again
Complete encode-process wall time (s)
181.01 s
170.62 s
168.69 s
33.66 s
Lossless
Repeat 1
Repeat 2
Repeat 3
Packing + lossless compression
Candidate decode + error checks
Scale + quantization
Global quality accounting
Inspection/preflight
Amplitude statistics
Exact columns + other overhead
Integrity/final publication
Visibility input reads
Gather blocks
Visibility payload writes
Sort IDs + group
Mean exclusive encode-function subregion (s)